CYBER SECURITY INTERNSHIP

TASK 5

Wireshark Packet Capturing and Network Traffic Analysis

Practical Report

Submitted by	**Harshika Choudhary**
Practical environment	Kali Linux running in VMware Workstation
Tool used	Wireshark
Capture interface	eth0

Task question / assignment

Capture and analyze network traffic using Wireshark. Capture live network packets, identify basic protocols and traffic types, apply protocol display filters, and summarize the packet details observed.

Objective

To capture live network packets on an active network interface and recognize the protocols present in the captured traffic.

Tools and environment

Wireshark (free packet analyzer), Kali Linux, VMware Workstation, and the terminal ping command to generate ICMP traffic.

Expected deliverables

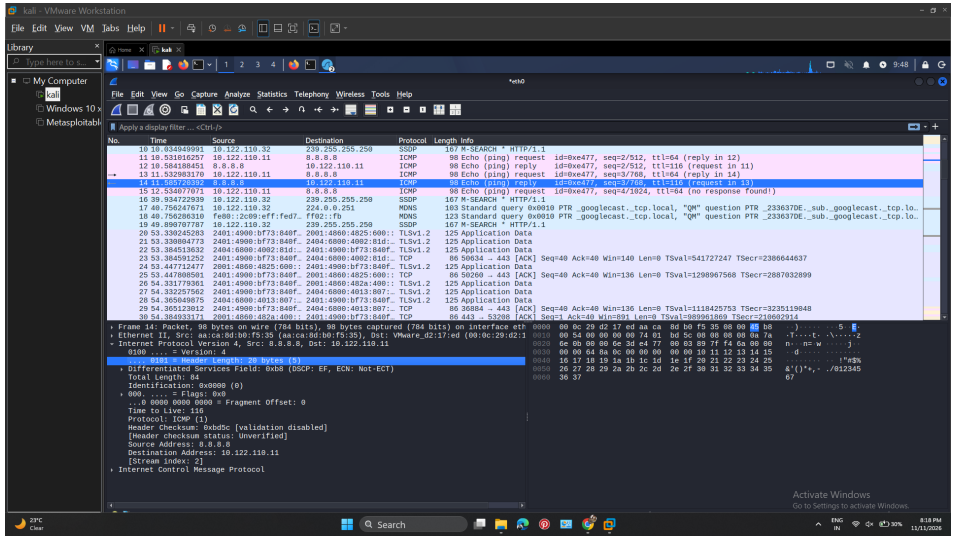
A packet capture file (.pcap) and a short report identifying at least three protocols. This report documents the visible results in the supplied screenshots. The .pcap file was not included with the screenshots and should be exported separately if required for submission.

Key concepts: packet capture • protocol identification • ICMP • TCP • TLS • network troubleshooting

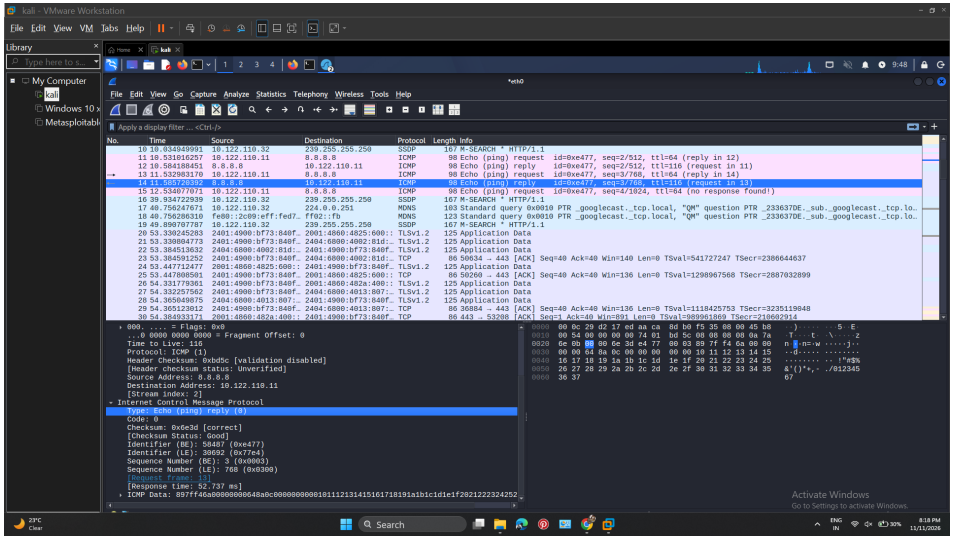
1. Practical procedure and live capture

The following steps describe the workflow followed for the practical. The screenshots show Wireshark capturing on the **eth0** interface in Kali Linux.

Step	Action
1	Open Wireshark in Kali Linux and select the active network interface (eth0).
2	Start the capture and generate network activity. In this practical, ping traffic to 8.8.8.8 was generated from the terminal.
3	Allow packets to appear in the packet list, then stop the capture after collecting sufficient traffic.
4	Inspect the Protocol, Source, Destination, Length and Info columns. Select individual packets to view their protocol fields and packet bytes.
5	Use display filters such as **icmp**, **tcp**, **tls**, **ssdp** or **mdns** to isolate protocol traffic. (The screenshots show protocol identification; they do not show the filter bar populated with these filters.)



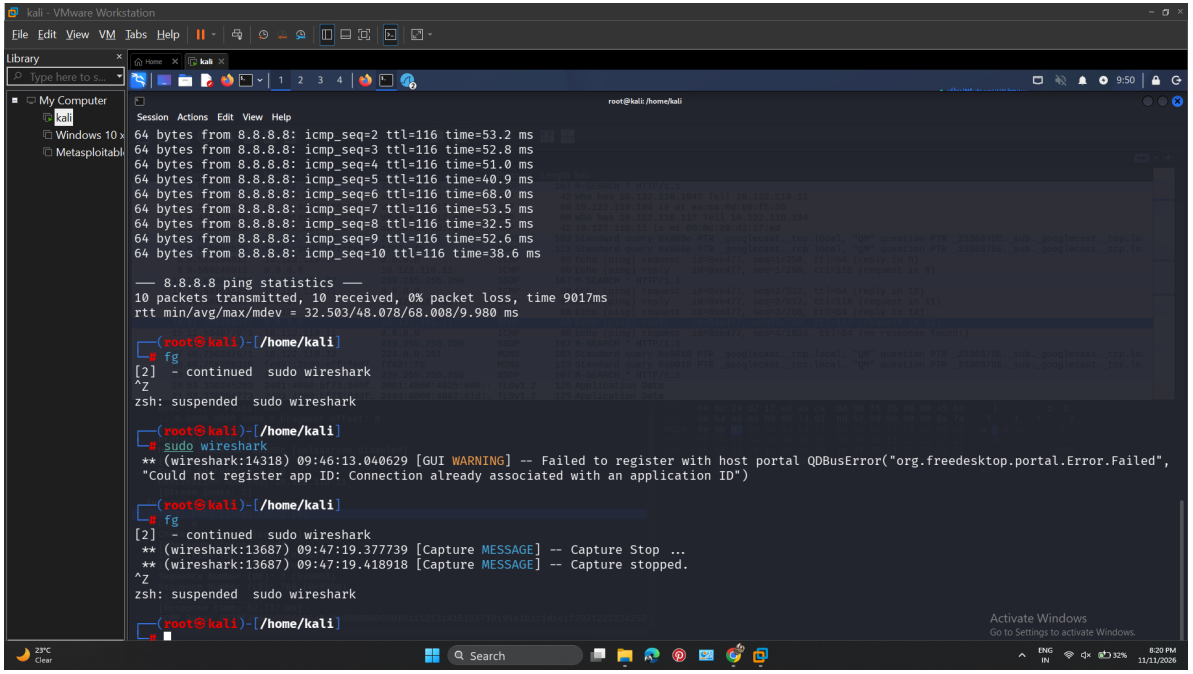
Screenshot 1. Wireshark packet list during live capture on eth0. Visible traffic includes ICMP, SSDP, mDNS, TLSv1.2 and TCP.



Screenshot 2. Selected ICMP Echo Reply packet. The details show source 8.8.8.8, destination 10.122.110.11, TTL 116 and ICMP Echo reply type.

2. Packet analysis, observations and result

The packet list in the supplied capture shows several protocol types. The selected packet is an ICMP Echo Reply from 8.8.8.8 to the Kali host (10.122.110.11). The terminal screenshot records a successful ping test.



Screenshot 3. Terminal output for ping to 8.8.8.8: 10 packets transmitted, 10 received, 0% packet loss; reported round-trip average 48.078 ms.

Protocols identified in the screenshots

Protocol	What it indicates in this capture
ICMP	Echo requests and replies associated with the ping test. The selected reply is from 8.8.8.8 to 10.122.110.11.
TCP	Transport-layer packets are visible, including traffic involving port 443.
TLSv1.2	Encrypted application-data records are visible. The screenshot labels these packets as TLSv1.2 Application Data.
SSDP	Local network service-discovery traffic is visible; the Info column shows M-SEARCH.
mDNS	Multicast DNS queries are visible for local service discovery, including _googlecast._tcp.local.

Result and conclusion

Live network traffic was captured and inspected successfully in Wireshark. At least three protocols were identified in the packet list: ICMP, TCP and TLSv1.2. SSDP and mDNS were also visible. The ping test received all 10 replies with no packet loss, and the selected ICMP packet details confirmed an Echo Reply. This practical demonstrates how Wireshark can be used to observe protocol activity and support basic network troubleshooting.

Submission note: Export the capture from Wireshark using File → Save As and select the pcap/pcapng format. Keep the capture file with this report in the Task 5 GitHub repository. The screenshots alone do not establish that a capture file was exported.